Classical Field Theory — Exercise Sheet 19

Adjoint identities, matter and surface currents

Solutions

All fields and coefficients on this sheet are smooth. Variations and gauge parameters in action identities have compact support inside the coordinate region. Use the Euler sign $\delta S = \int E_a \delta\phi^a$ and the connection conventions of Lecture 19. No equations of motion are imposed unless explicitly stated.

Exercise 19.1 (A second-order adjoint and the elementary converse). On an open coordinate patch $U \subset \mathbb{R}^{1+1}$ with coordinates (t, x), let a scalar parameter $\varepsilon \in C_c^\infty(U)$ produce two variations

$$R(\varepsilon)^1 = (1+tx)\varepsilon_{tt} + 2(t+x^2)\varepsilon_{tx} + e^t\varepsilon_x, \qquad R(\varepsilon)^2 = (1+t^2)\varepsilon_{xx} + x\varepsilon.$$

Compute $R^*(E_1, E_2)$, first in divergence form and then with every coefficient derivative expanded. Prove the compact pairing identity. Suppose an action has the compact first variation $\int E_a[\phi]h^a$ and $R^*_\phi E[\phi] = 0$ on U for every smooth field. Prove that $R_\phi(\varepsilon)$ is an infinitesimal gauge symmetry. Explain why the same converse holds for any finite-order operator $R_\phi(\varepsilon)^a = \sum_I r^{aI}_\alpha[\phi]\partial_I\varepsilon^\alpha$, without interpreting it as a converse for arbitrary conserved currents.

Solution. Repeated coordinate integration by parts gives

$$R^*E = \partial_t^2[(1+tx)E_1] + 2\partial_t\partial_x[(t+x^2)E_1] - \partial_x(e^tE_1) + \partial_x^2[(1+t^2)E_2] + xE_2.$$

For example the mixed term expands as $2(t+x^2)E_{1,tx} + 2E_{1,x} + 4xE_{1,t}$. The first term adds $(1+tx)E_{1,tt} + 2xE_{1,t}$. The other terms therefore yield the full expression

$$R^*E = (1+tx)E_{1,tt} + 2(t+x^2)E_{1,tx} + 6xE_{1,t} + (2-e^t)E_{1,x} + (1+t^2)E_{2,xx} + xE_2.$$

Each discarded derivative contains a derivative of the compact parameter. Extend by zero and use Fubini/FTC to remove it. This proves $\int E_a R(\varepsilon)^a = \int (R^*E)\varepsilon$ without a boundary theorem.

For the converse, apply the supplied first variation to the compact variation $h = R_\phi(\varepsilon)$. At the fixed field its coefficients are smooth, so this variation is still compactly supported in U. The pairing identity and the assumed off-shell identity give

$$\delta S[\phi]R_\phi(\varepsilon) = \int (R^*_\phi E[\phi])\varepsilon = 0.$$

The same finite integrations by parts give $(R^*_\phi E)_\alpha = \sum_I (-1)^{|I|}\partial_I(r^{aI}_\alpha E_a)$ for arbitrary finite order. Hence the same conclusion holds on U. Holding the field fixed does not freeze the evaluated coefficients under coordinate differentiation. No assertion that an arbitrary conserved current comes from a symmetry was used or proved.

Exercise 19.2 (The Euler term carried by charged matter). Write $A_{\mathrm{em}} = A_\mu \mathrm{d}x^\mu$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and let $u = (u_1, u_2)^{\mathsf{T}}$ be two real scalar fields. Given $q, \lambda, v > 0$, set

$$T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D_\mu u = \partial_\mu u - q A_\mu T u, \qquad L = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} D_\mu u \cdot D^\mu u - \frac{\lambda}{8}(|u|^2 - v^2)^2.$$

The gauge variations are $\delta A_\mu = \partial_\mu \varepsilon$ and $\delta u = q\varepsilon T u$.

(a) Verify covariance of $D_\mu u$, derive both Euler arrays by compact variation, and derive the off-shell Noether identity containing them.

(b) On a coordinate patch take $A_\mu = 0$ and $u = (t, t^2)^{\mathsf{T}}$. Compute the ordinary divergence of the total gauge Euler array and its balancing scalar Euler term. Does the pure Maxwell identity remain an identity for that total array by itself?

(c) For a real orthogonal representation with skew generators t_a, use instead the supplied variations $\delta A_\mu = D_\mu \varepsilon$, $\delta u = -\varepsilon^a t_a u$. Assume these variations are a gauge symmetry for every compact parameter. If the compact first variation is $\int [\langle E_A^\mu, \delta A_\mu \rangle + E_u \cdot \delta u]$, derive its covariant component identity. Here A is a connection as in Lecture 19, and the Lie algebra basis is orthonormal in its invariant pairing.

Solution. For the first part, direct differentiation cancels $q(\partial_\mu \varepsilon) T u$ against the potential variation and leaves $\delta(D_\mu u) = q\varepsilon T(D_\mu u)$. Skewness of T and radiality of the potential show $\delta L = 0$. The covariant product rule, with $T^{\mathsf{T}} = -T$, permits compact integration by parts. It gives

$$E_u = -D_\mu D^\mu u - \frac{\lambda}{2}(|u|^2 - v^2)u, \qquad E_A^\mu = \partial_\nu F^{\nu\mu} - q(Tu) \cdot D^\mu u.$$

In particular the matter contribution to E_A^μ has a minus sign, because $\delta_A D_\mu u = -q\delta A_\mu T u$. Substituting the gauge variation into the first variation yields

$$0 = \int \left[E_A^\mu \partial_\mu \varepsilon + q E_u \cdot Tu\, \varepsilon \right] = \int \left[-\partial_\mu E_A^\mu + q E_u \cdot Tu \right] \varepsilon.$$

The compact smooth-test fundamental lemma proves the local off-shell identity.

For the second part, $Tu = (-t^2, t)^{\mathsf{T}}$ and $(Tu) \cdot u_t = t^2$. All spatial derivatives and the curvature vanish. Thus

$$E_A^0 = -qt^2, \quad E_A^i = 0, \quad \partial_\mu E_A^\mu = -2qt.$$

The radial potential term is orthogonal to Tu, while $-u_{tt} \cdot Tu = -(0, 2) \cdot (-t^2, t) = -2t$. Hence $qE_u \cdot Tu = -2qt$, exactly balancing the divergence. Except at $t = 0$, that divergence is nonzero. The identity $\partial_\mu \partial_\nu F^{\nu\mu} = 0$ still holds for the pure curvature part, but it does not annihilate the total matter-coupled Euler array.

For the final part, invariance and compact integration by parts give

$$0 = \int \left[-(D_\mu E_A^\mu)^a - E_u \cdot t_a u \right] \varepsilon^a.$$

The fundamental lemma proves $(D_\mu E_A^\mu)^a + E_u \cdot t_a u = 0$. This sign uses the explicitly supplied connection variation and matter variation; it must not be guessed by replacing an ordinary derivative in the abelian formula without also translating its coupling convention.

Exercise 19.3 (Noncommuting constant connections). Let $T_a = -\mathrm{i}\sigma_a/2$, so $[T_a, T_b] = \epsilon_{abc} T_c$. In pure Yang–Mills theory with coupling $g > 0$, take first $A_1 = T_1$, $A_2 = T_2$, $A_0 = A_3 = 0$, all constant. Compute every nonzero curvature and Euler component $E^\nu = g^{-2} D_\mu F^{\mu\nu}$, and check the covariant identity. Repeat after changing A_3 to $T_1 + T_2$. Exhibit the separate nonzero commutator contributions to $D_\nu E^\nu$ in that second case. What can these constant examples establish about an ordinary-divergence identity for arbitrary nonconstant connections?

Solution. For the first connection the only independent curvature is $F_{12} = [T_1, T_2] = T_3$, with $F_{21} = -T_3$. All temporal curvature and E^0 vanish. Spatial raising gives $F^{ij} = F_{ij}$, so

$$E^1 = g^{-2}[T_2, -T_3] = -g^{-2}T_1, \qquad E^2 = g^{-2}[T_1, T_3] = -g^{-2}T_2, \qquad E^3 = 0.$$

Each $[A_i, E^i]$ vanishes in this first case. For the second connection the independent curvatures are

$$F_{12} = T_3, \qquad F_{13} = T_3, \qquad F_{23} = -T_3,$$

with the reversed-index negatives and all temporal entries zero. Summing the two relevant commutators for each spatial Euler component gives

$$E^1 = g^{-2}(-2T_1 + T_2), \quad E^2 = g^{-2}(T_1 - 2T_2), \quad E^3 = -g^{-2}(T_1 + T_2), \quad E^0 = 0.$$

Their ordinary derivatives vanish because the fields are constant, but

$$[A_1, E^1] = g^{-2}T_3, \qquad [A_2, E^2] = -g^{-2}T_3, \qquad [A_3, E^3] = 0.$$

The nonzero terms cancel, verifying the covariant identity without erasing its commutators separately. Ordinary divergence happens to vanish for these constant fields. That observation cannot prove an off-shell ordinary-divergence identity for general connections; Lecture 19's variable-connection guard exhibits why the covariant derivative is essential.

Exercise 19.4 (A local superpotential theorem and a Maxwell verification). For this exercise, assume the following local decomposition statement. It is an additional hypothesis here, not a theorem to infer from the fundamental lemma alone. For a finite-order local Lagrangian of unconstrained $\mathbb{R}^N$-valued fields on a star-shaped coordinate patch of spacetime dimension at least two, a finite-order infinitesimal gauge symmetry has a local Noether current of the form

$$j^\mu_\varepsilon = W^\mu_\varepsilon + \partial_\nu U^{\nu\mu}_\varepsilon, \qquad U^{\nu\mu}_\varepsilon = -U^{\mu\nu}_\varepsilon.$$

The current and these expressions depend linearly on the parameter and finitely many of its derivatives, with smooth finite-order local field coefficients. The remainder W vanishes when the Euler expressions and their derivatives vanish. This is a local statement on the full jet space; no global topological superpotential theorem is supplied.

Verify it directly for

$$L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + A_\mu j^\mu_{\text{ext}}, \qquad \partial_\mu j^\mu_{\text{ext}} = 0, \qquad \delta A_\mu = \partial_\mu \varepsilon,$$

where the external current is fixed and smooth. Derive the gauge Noether current, its Euler remainder and an explicit U. On solutions, compare its spatial integral for compact spatial parameters with a parameter equal to a constant ε_∞ outside a ball. For the latter comparison assume a smooth Maxwell field whose external current has compact spatial support on the slice and total charge $Q = \int j^0_{\text{ext}} \mathrm{d}^3x$. Use $F_{i0} = E_i$ and an outward sphere orientation to find the surface sign. No Schwartz or L^2 condition on the potential is imposed in this boundary comparison.

Solution. The source conservation law gives $\delta L = j^\mu_{\text{ext}}\partial_\mu \varepsilon = \partial_\mu(\varepsilon j^\mu_{\text{ext}})$. The momentum derivative with respect to $\partial_\mu A_\nu$ is $-F^{\mu\nu}$, so Lecture 2's formula gives

$$j^\mu_\varepsilon = -F^{\mu\nu}\partial_\nu \varepsilon - \varepsilon j^\mu_{\text{ext}}, \qquad E^\mu_A = \partial_\nu F^{\nu\mu} + j^\mu_{\text{ext}}.$$

Antisymmetry of F and the ordinary product rule give the exact off-shell decomposition

$$j^\mu_\varepsilon = -\varepsilon E^\mu_A + \partial_\nu U^{\nu\mu}_\varepsilon, \qquad U^{\nu\mu}_\varepsilon = \varepsilon F^{\nu\mu}.$$

This verifies the assumed decomposition here with an Euler remainder, by direct calculation. The background source need not satisfy a field equation; its prescribed conservation is what ensured gauge invariance up to a divergence.

On a solution the time-slice density is a spatial divergence, since $U^{00} = 0$. For a compact spatial parameter, U^{i0} vanishes outside a compact set, so coordinate cutoff integration gives $\int j^0_\varepsilon = 0$. For a parameter constant outside a ball containing the source support, its current density is still compactly supported: the parameter derivatives and the source term vanish outside that ball. Using Lecture 5's ball formula, for every sufficiently large sphere,

$$\int j^0_\varepsilon \mathrm{d}^3x = \int_{S^2_r} \varepsilon_\infty F^{i0} n_i \mathrm{d}S = -\varepsilon_\infty \int_{S^2_r} \mathbf{E}\cdot n \mathrm{d}S = -\varepsilon_\infty Q.$$

The minus sign comes from $F^{i0} = -E_i$. The electric Gauss charge itself is $Q = +\int \mathbf{E}\cdot n$; its sign was not changed. Compact-parameter identities cannot discard the surface term of a parameter with a nonzero limit. The explicit Maxwell calculation establishes this boundary comparison without claiming that the assumed local statement solves a global gauge or topological problem.